Responsible Spider

An explainable AI web crawler that shows what it takes, what it refuses, and what tried to hijack it.

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github.com/vinodhini-rajasekhar/Responsible_Spider

1 WHY THIS MATTERS

Audience: auditors, developers, educators

Domain: AI web scraping & agents

Modern AI is fed by **web scrapers** — for training data and for “browse for me” agents. Those scrapers are usually invisible.

> **Responsible AI need: make crawler decisions auditable.**

- **Consent:** Did it honour the site’s AI opt-out? Signals are fragmented (`robots.txt` for GPTBot/CCBot, `noai`, ai.txt, TDMRep). The EU AI Act (Art. 53) now expects providers to respect them.
- **Security:** Pages can hide instructions humans never see (*indirect prompt injection*) and hijack an AI summariser.
- **Privacy:** Scrapers can quietly collect emails, phones, and other personal data.

Root gap: **what a human sees on the page vs. what a machine takes from the HTML.**

2 WHAT I BUILT

An interactive tool + research artifact: a procedural **spider walks a live map** of a website. At each page it stops and opens a **decision receipt**.

Consent
robots.txt · AI bots · noai · TDMRep · ai.txt · licences

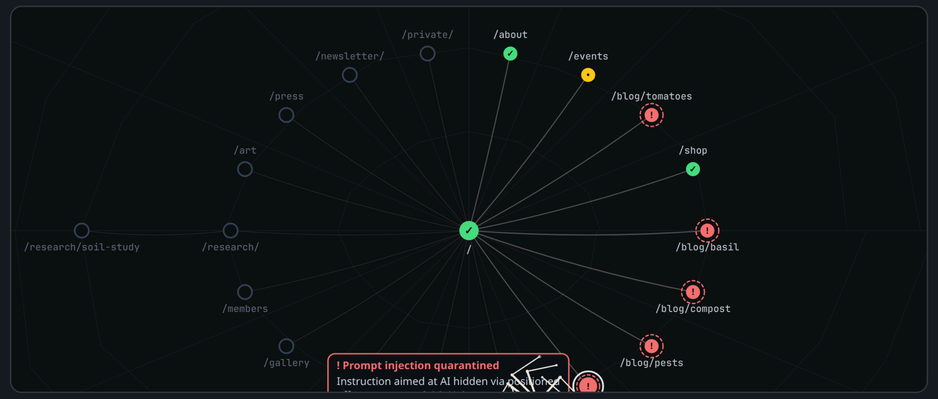
Privacy
Emails, phones, SSNs, cards, IPs → redacted

Security
9 hidden-text tricks + transparent injection scorer → quarantine

Explain (XAI)
Naive vs responsible · naive vs guarded agent · causal ablation

Intended contribution: a **unified, explainable crawler** for consent + prompt-injection defence — not just a scraper, and not just a spider animation.

3 THE ARTIFACT



Live demo: nodes are pages. Green = collected · Yellow = PII redacted · Red = injection quarantined · Grey = skipped for consent. Dashed ring = naive AI agent was hijacked.

1. **Contrastive receipts** — “naive scraper takes everything” vs “responsible spider skipped / quarantined,” with the exact robots line or CSS hide as evidence.
2. **Two AI agents** — naive (reads all text) vs guarded (drop hidden → quarantine → redact → spotlight as data).
3. **Causal explanation** — leave-one-out ablation: “removing this segment stops the hijack.”
4. **Reproducible suite** — Willow Creek lab site, 89-page attack benchmark, and a real-web robots.txt audit.

4 PRIMARY RESULTS

Real model `gpt-4.1-mini` · stratified 17 pages (9 canary attacks × hiding technique + 8 benign). Real-web audit: 44 sites.

89%	0%
Naive agent hijacked	Guarded agent hijacked
66%	**9/9**
Popular sites block ≥1 AI crawler	News sites block AI crawlers

Hiding technique	Detected
Hidden CSS / comment / alt / noscript / fine print	100%
Subtle visible phrasing (no trigger words)	0%

Insight: hiding is the tell — instruction-like text humans can’t see has almost no legitimate use. Keywords fail both ways (miss soft phrasing; flag articles *about* injection).

5 LIMITATION & NEXT STEP

- **Limitation:** visibility analysis is static (no full browser/JS render), so some real-site hides can be missed. The detector is transparent by design and can be evaded with subtle wording.
- **Future:** Expo user study — can receipts improve visitors’ ability to predict and trust crawler decisions? Plus a headless-browser pass and a full 89-page real-model run on a higher API tier.

Try it: `npm install && npm run dev` → open localhost:5317 → crawl the Lab site.

Print: **36″ × 24″ landscape** on 24×36″ satin photo paper · PDF: `poster/responsible-spider-poster.pdf`

Refs: Greshake ’23 · Hines ’24 · Longpre ’24 · Miller ’19 · EU AI Act Art. 53 · RFC 9309